# AROB Reinforcement Learning Report

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## Introduction

The goal of this report is to evaluate and compare the Q-learning (model free) and Dyna-Q (model based) tabular reinforcement learning algorithms on the deterministic maze environment of varying sizes. Specifically, we aim to compare their sample and time efficiency.

## Methods

### Environment

We work on the Maze MDP problem for mazes of size ($3 \times 3$, $5 \times 5$ and $8 \times 8$). The goal in this environment is for the agent, starting in the top left corner of the maze, to find the optimal path to the reward found at the bottom right of the maze. In order to make our problem fair for all environment sizes, the obstacles were removed.

### Algorithms

Q-learning is a model-free algorithm that iteratively approaches the optimal Q-values for all states using local updates. To perform a Bellman backup, the temporal difference error $\delta_t$ is computed and used to update $Q$ as:

$$\delta_t = \left( r(s_t, a_t) + \gamma \max_{a \in \mathcal{A}} Q^{(i)}(s_{t+1}, a) \right) - Q^{(i)}(s_t, a_t)$$

$$Q^{(i+1)}(s_t, a_t) = Q^{(i)}(s_t, a_t) + \alpha \delta_t$$

Dyna-Q is a model-based reinforcement learning algorithm that builds a model of the environment and uses it to simulate experience from the environment "in imagination". The environment model is comprised of transition, reward and termination models that are chosen to be deterministic. It uses the same temporal difference update as Q-learning, according to Richard Sutton's implementation, applies it to both transitions from the environment and transitions generated from the model, however our implementation Bellman backups are applied only for the samples from the model. The goal is to improve learning efficiency by being able to find the optimal policy with less sample transitions from the environment.

Exploration policies for both Q-learning and Dyna-Q are using softmax function based on current Q-values to sample the next action.

### Hyperparameter Tuning

For both Q-learning and Dyna-Q, using Optuna and looking at algorithm convergence we found that the optimal parameters for learning rate was $\alpha = 1.0$. We found that setting the learning rate $\alpha \geq 1.1$ would result in the norm of Q diverging.

When it comes to the temperature parameter $\beta$ Optuna generally found optimal values for Q-learning and Dyna-Q between 3 and 5. We ended up choosing the middle value of $\beta = 4$. However, we note that for both Q-learning and Dyna-Q we found that generally the temperature between $\beta = 2$ and $\beta = 12$ had no consistent effect on performance.

Discount factor $\gamma$ is considered as part of the environment and is equal to 0.9.

### Evaluation Metrics

In order to evaluate the time and sample efficiency, we measured the joint evolution of Q value norms with computation time and with number of used samples for both algorithms across three maze sizes. The number of used samples corresponds to number of actuals steps in the environment, excluding "in imagination" steps in case of Dyna-Q. Since time and number of samples values are not fixed and we want to compare the general evolution of Q value norms, we have chosen the area under the curve (AUC) as performance metric. Higher AUC values correspond to higher time or sample efficiency under assumption that Q value norm monotonically increases towards the optimum, which is the case for all individual curves (Figures 1, 2, 3). In order to compare AUC between algorithms, the same time or sample interval was considered. Q-value norms used for computing AUC were scaled by optimal Q value norm obtained with policy iteration algorithm, so that we can compare performances across the environments as well.

### Statistical Test

Each experiment was repeated 100 times and then mean AUC values were compared using Welsh's t-test. The null hypothesis of Welsh's t-test is that two distributions have the same mean. We considered p-value significance thresholds of 0.05,0.01 and 0.001.

## Results and Discussion

### Time Efficiency

As can be seen in Figures 1 and 2, Q-learning shows a faster convergence than Dyna-Q for $3 \times 3$ and $5 \times 5$ mazes, which is confirmed by AUC comparisons with the Welsh t-test, both of which yield p-values below 0.001. However, this trend is opposite for the $8 \times 8$ maze (Figure 3). Dyna-Q, although initially shows slower progression in Q value norm, eventually takes over and converges faster than Q-learning; this is supported by statistically significant ($p < 0.01$) difference between AUC values. 4. We can thus conclude that for mazes of small size Q-learning is more time efficient, likely due to the fact that it quickly finds the goal and efficiently propagates the reward, whereas Dyna-Q spends extra computation time for modeling a simple environment. However, for larger $8 \times 8$ maze, Dyna-Q shows better time efficiency, we can suppose that it is due to more efficient reward propagation with "in imagination" learning.

### Sample Efficiency

We can see from Figures 1, 2 and 3 that Dyna-Q shows faster learning as a function of the number of transition samples the algorithm has seen. This is confirmed by comparison of AUC means (Figure 4), showing statistically significant difference with p-values of less than 0.001.

The explanation for this again comes down to the "in imagination" learning that the Dyna-Q does. By learning from simulated transitions generated by its model, as long as the model is good enough, for each sample transition from the environment it takes it can take multiple steps "in imagination". This increases the rate of convergence with respect to the number of environment transitions.

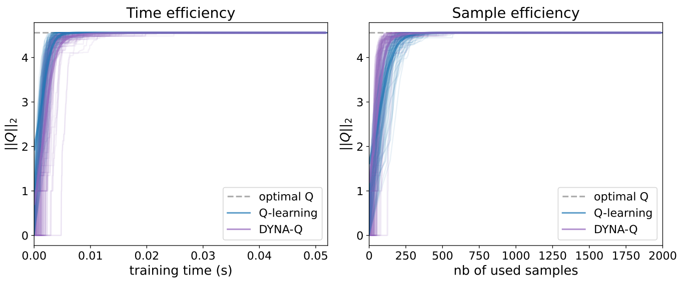


Figure 1: Time and sample efficiency of Q-learning and Dyna-Q for 3×3 maze. Both algorithms show rapid Q value learning both in terms of computation time and of environment samples. Opaque curves show the joint evolution of Q value norm and training time or number of used samples for each of 100 algorithm runs. Curves in bold show rolling average for corresponding curves, with slight artifacts at the edges due to padding. Gray dashed line shows the Q-value norm obtained with policy iteration and treated as the reference level.

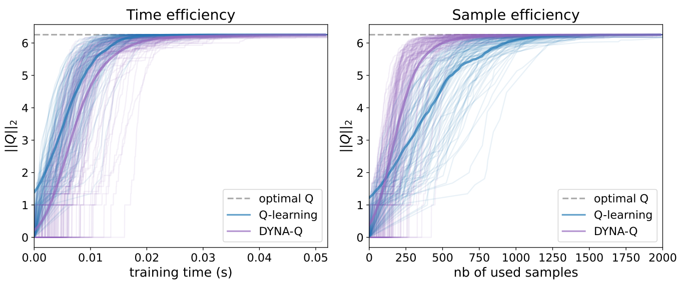


Figure 2: Time and sample efficiency of Q-learning and Dyna-Q for $5 \times 5$ maze. Both algorithms show moderate Q value learning both in terms of computation time and of environment samples. We can notice that some of learning curves for Dyna-Q are initially stuck at low Q values. Same conventions as in Figure 1.

### MDP Size Effect

Q value norm evolutions on Figures 1, 2 and 3 as well as comparisons of AUCs on Figure 4 show statistically significant differences in time and sample efficiency between mazes for both algorithms ($p < 0.001$ for all comparisons, not shown on plot). It is consistent with the fact that larger mazes require more time and more environment steps (samples) to be explored. As noted above, the effect of maze size on the time efficiency of Q-learning is stronger than on that of Dyna-Q. However, closer examination of individual curves on Figure 3 shows that Dyna-Q is more prone to be initially blocked at low Q values, which suggests that the model learned initially was not good enough to correctly propagate reward "in imagination".

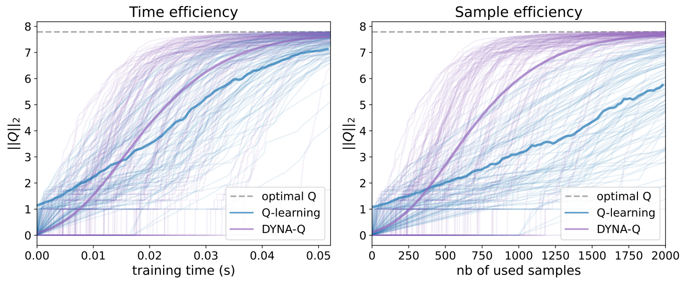


Figure 3: Time and sample efficiency of Q-learning and Dyna-Q for $8 \times 8$ maze. Both algorithms show rather slow Q value learning both in terms of computation time and of environment samples. During initial learning steps, substantial number of DynaQ runs are stuck at low Q values. Same conventions as in Figure 1.

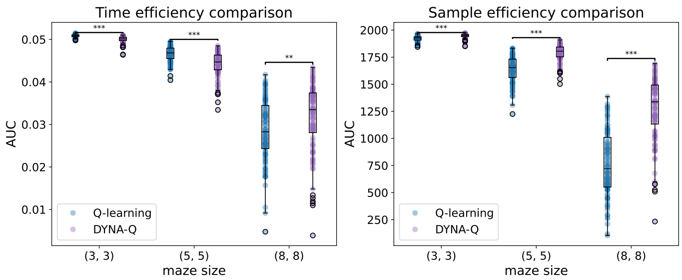


Figure 4: Time efficiency and sample efficiency comparison between Q-learning and Dyna-Q across maze sizes. Area under normalized performance curves (AUC) on the same time and sample interval were taken. Dyna-Q is significantly less time efficient and more sample efficient across all mazes, except for being more time efficient for $8 \times 8$ maze. Larger environments are associated with higher computation time and require more environment steps. Boxplots depict the median, first and third quartiles, as well as outliers above 1.5 inter-quartile range from closest quartiles. Corresponding means were compared with Welsh t-test, significance levels of p-values are $\star$ for 0.05, $\star\star$ for 0.01, and $\star\star\star$ for 0.001. P-values for mean comparisons between mazes are not shown and all are below 0.001.

## Conclusion

Experiments on maze environments on sizes $3 \times 3$, $5 \times 5$ and $8 \times 8$ show that both models are able to solve the maze and find the optimal solution. Compared to Q-learning, Dyna-Q is more sample efficient for all environments, less time efficient for small ($3 \times 3$ and $5 \times 5$) and more time efficient for larger ($8 \times 8$) environments. Our results suggest that model-based learning is more sample and time efficient for complex environments, however in simple cases model free approach is enough and is even more time efficient. Additionally, the potential trade-off is that the model based algorithm can be initially stuck because of poorly learned model. The experiments on other more complex environments are required to explore further this trade-off.